

Schematic Sheets

#	Name	File	Description
1	Main	SensWear_Main.SchDoc	Top-level schematic showing system hierarchy and interconnection between components
2	MCU	SensWear_MCU.SchDoc	Micro-Controller Unit (MCU)
3	Memory	SensWear_Memory.SchDoc	External EEPROM / non-volatile memory
4	Power	SensWear_Power.SchDoc	Battery input, voltage regulation, charging circuitry, and daughter board regulator
5	MPU	SensWear_MPU.SchDoc	Motion sensing section (IMU)
6	LED	SensWear_RGB.SchDoc	RGB/status LED circuitry
7	Debug	SensWear_Debug.SchDoc	Debug and development interface including UART and SWD/programming access
8	DaughterBoard	SensWear_DaughterBoard.SchDoc	Modular daughter-board interface exposing power, communication buses, and GPIOs

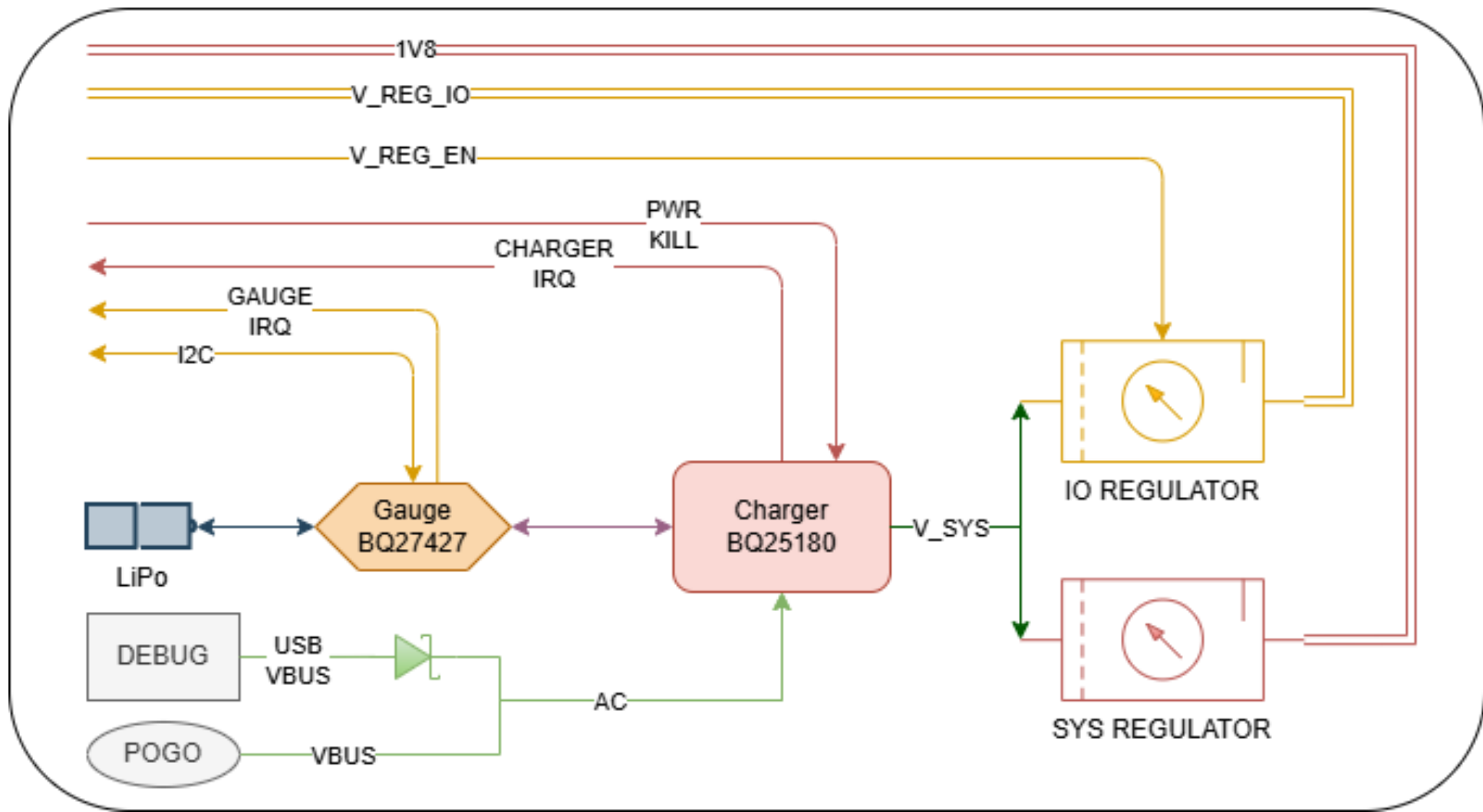
Revision History

Revision	Date	Description
A0	May 2025	Initial validation design on rigid PCB
B0	March 2026	Flexible PCB manufacturing updates



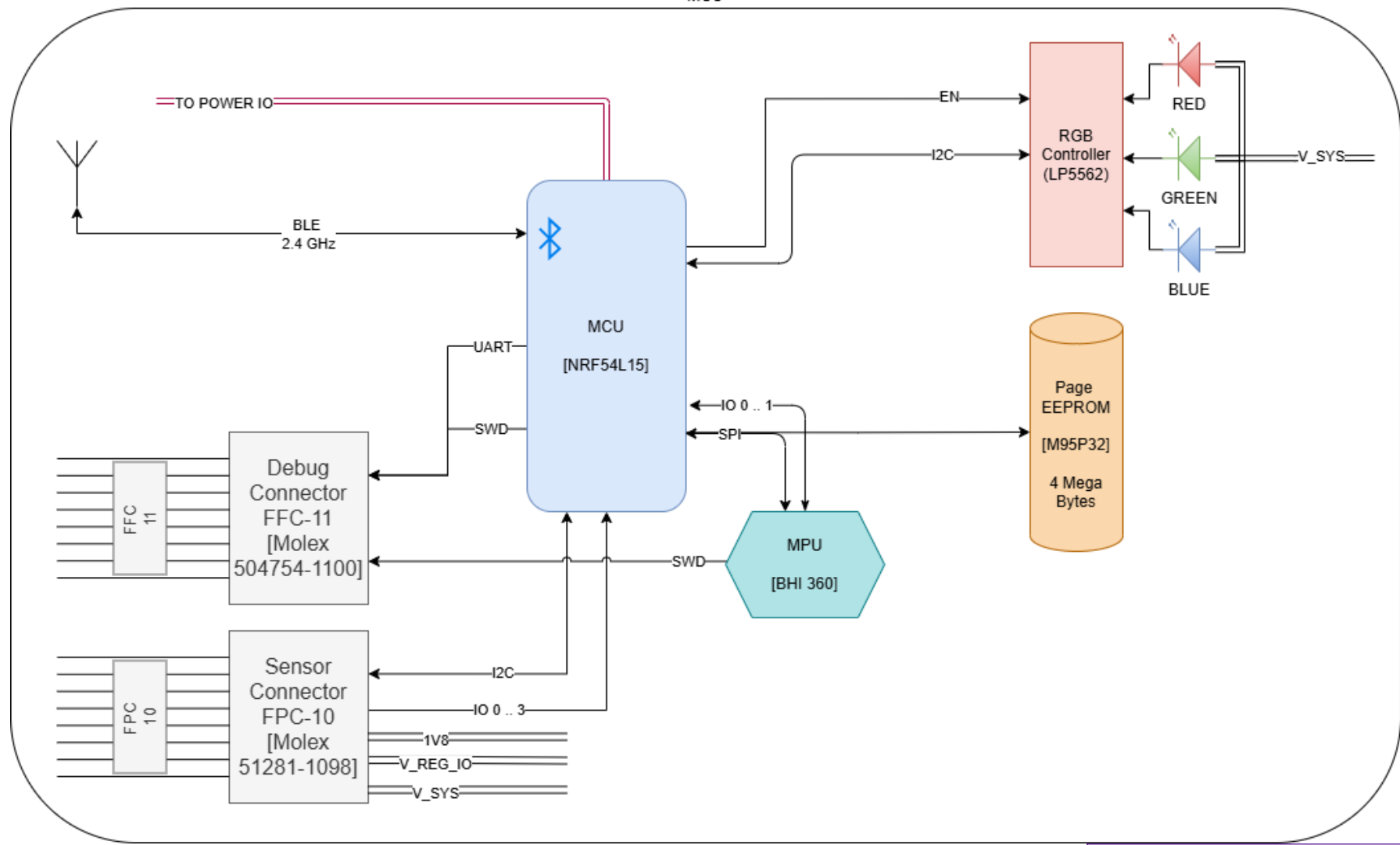
SensWear-V1R1

Designer/Engineer	Board Title		
Yusein Ali	SensWear-Hub		
Drawn By	Document Name		
Yusein Ali	SensWear-V1R1-Hub-		
Approved By	Ver.	Rev.	Date
Dariush Salami	1	B0	01.03.2026



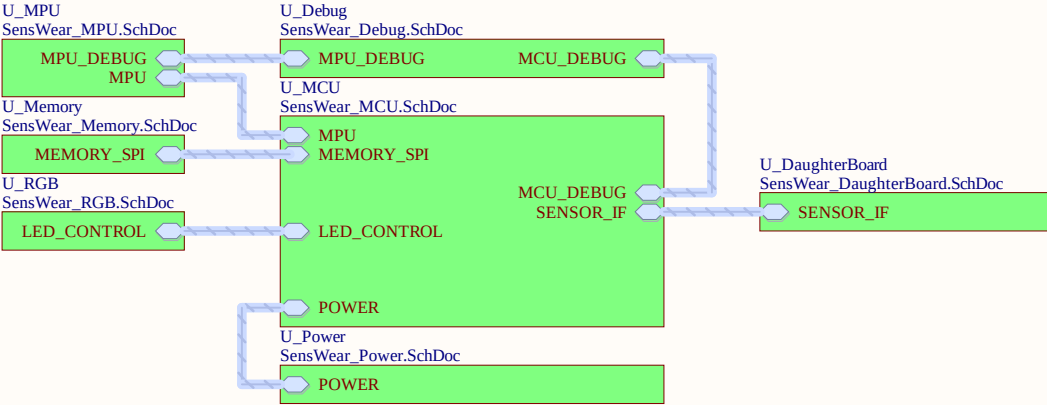
SensWear-V1R1

Designer/Engineer	Board Title		
Yusein Ali	SensWear-Hub		
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Yusein Ali	SensWear-V1R1-Hub-		
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Designer/Engineer	Board Title		
Yusein Ali	SensWear-Hub		
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Yusein Ali	SensWear-V1R1-Hub-		
Approved By	Ver.	Rev.	Date
Dariush Salami	1	B0	01.03.2026



		SensWear-V1R1			
Designer/Engineer Yusein Ali		Board Title SensWear-Hub			
Drawn By Yusein Ali		Document Name SensWear-Main.SchDoc			
Approved By Dariush Salami	Rev. B0	Sheet	1	of	8
		Date	01.03.2026		
Size		A4			

nRF54L15 PERIPHERAL PIN ASSIGNMENTS

[UART] - Instance: UART30 (Peripheral Domain)
 RXD : P0.03 (Input)
 TXD : P0.04 (Output)
 NOTE: P0.03 and P0.04 are valid P0 digital I/O pins and should be used with
 UARTE30.

[SPI MASTER] - Instance: SPIM00 (MCU Domain - High Speed)
 SCK : P2.01
 MOSI : P2.02 (Dedicated IO0 Pad)
 MISO : P2.04 (Dedicated IO1 Pad)
 NOTE : These pins utilize dedicated hardware muxing.
 Do not swap MOSI/MISO on these pins.

[I2C MASTER] - Instance: TWIM20 (Peripheral Domain)
SDA : P1.02
SCL : P1.03
REQ : External Pull-up Resistors required.

I2C ADDRESS TABLE (I2C21)

Device Type R/W=x)	Part Number	Hex (7-bit)	Binary (8-bit with R/W=x)
Buck Converter	TPSM831021	0x68	0100 001x
Touch Controller	MTCH6102	0x25	0100 101x
LED Driver	LP5562	0x30	0110 000x
Temp Sensor	MAX30208	0x50	1010 000x
Fuel Gauge	BQ27427	0x55	1010 101x
Pulse Oximeter	MAX30101	0x57	1010 111x
Haptic Driver	DRV2605L	0x5A	1011 010x
Battery Charger	BQ25180	0x6A	1101 010x

**NOTE: 'x' = 0 for WRITE, 1 for READ.
Ensure 1.5k pull-ups on SDA/SCL lines.**

[CRYSTAL X2 NOTE]

Type: 32.768 kHz (CL=12.5pF)
Caps: INTERNAL (Firmware Enabled).
Do NOT place external capacitors.

```
&lfxo {
    status = "okay";
    load-capacitors = "internal";
    /* Calculation for 12.5pF Crystal:
    * Cpin = 2 * (Cload - Cstray)
    * Cpin = 2 * (12.5pF - 4pF) = 17pF
    */
    load-capacitance-femtofarad = <17000>;
};
```

SENSOR INTERFACE (GENERIC)

Logic Level: 1.8V (V_REG_IO Domain)

[SHARED CONTROL BUS]

Interface : I2C (I2C21)
SDA : P1.02 (Pin 3)
SCL : P1.03 (Pin 4)
Pull-ups : 1.5kΩ to 1V8 (R15, R16) installed on board.
Note : P1.02 and P1.03 are NFC pins by default.
Disable NFC since these pins are used as I2C
GPIO.

[SENSOR GPIOs / ANALOG INPUTS]

GPIO_0 : P1.14 / AIN7 (Pin 42)
- Capable of Analog Input (SAADC).
GPIO_1 : P1.13 / AIN6 (Pin 41)
- Capable of Analog Input (SAADC).
GPIO_2 : P1.12 / AIN5 (Pin 40)
- Capable of Analog Input (SAADC).
GPIO_3 : P2.11 / AIN4 (Pin 39)
- Capable of Analog input (SAADC).

[POWER DOMAIN]

Supply : V_REG_IO (Switched Rail)
Control : V_REG_EN (P1.06) must be HIGH.
Note : Ensure sensors connected to these GPIOs are powered by V_REG_IO to avoid leakage currents when the sensor subsystem is OFF.

MCU (nRF54L15)DEBUG & DIAGNOSTIC INTERFACE
Logic Level: 1.8V (VDD_IO)

[SWD PROGRAMMING]

- SWDIO : SWDIO (Physical Pin 25)
 - Bidirectional Data line.
 - Has internal pull-up.
- SWDCLK : SWDCLK (Physical Pin 26)
 - Clock input from debugger.
 - Has internal pull-down.
- RESET# : nRESET (Physical Pin 30)
 - Active Low System Reset.
 - Mandatory for device recovery (un-bricking).

[UART CONSOLE] - Instance: UART30

MCU_RX : P0.03 (Pin 28)
- Input to nRF54. Connect to Debugger TX.

MCU_TX : P0.04 (Pin 29)
- Output from nRF54. Connect to Debugger RX.

NOTE:
P0.03 also has GRTC PWM functionality.
P0.04 also has GRTC CLKOUT32K functionality.
Do not enable those GRTC functions while using these pins for UART.

[LAYOUT REQUIREMENTS]

1. Debug connector VREF/VTref must be connected to 1V8.
2. Debug probe / USB-UART must support 1.8V target I/O.
3. Do not place large capacitors on SWDIO or SWDCLK.
4. Keep SWDIO, SWDCLK, RESET#, UART_TX, and UART_RX short and clean.
5. UART_TX and UART_RX are single-ended signals, not differential pairs.
6. Provide a solid common GND between the debug board and the target board.

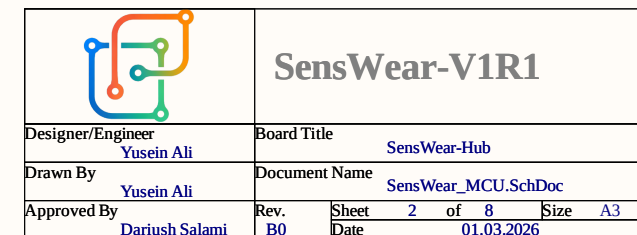
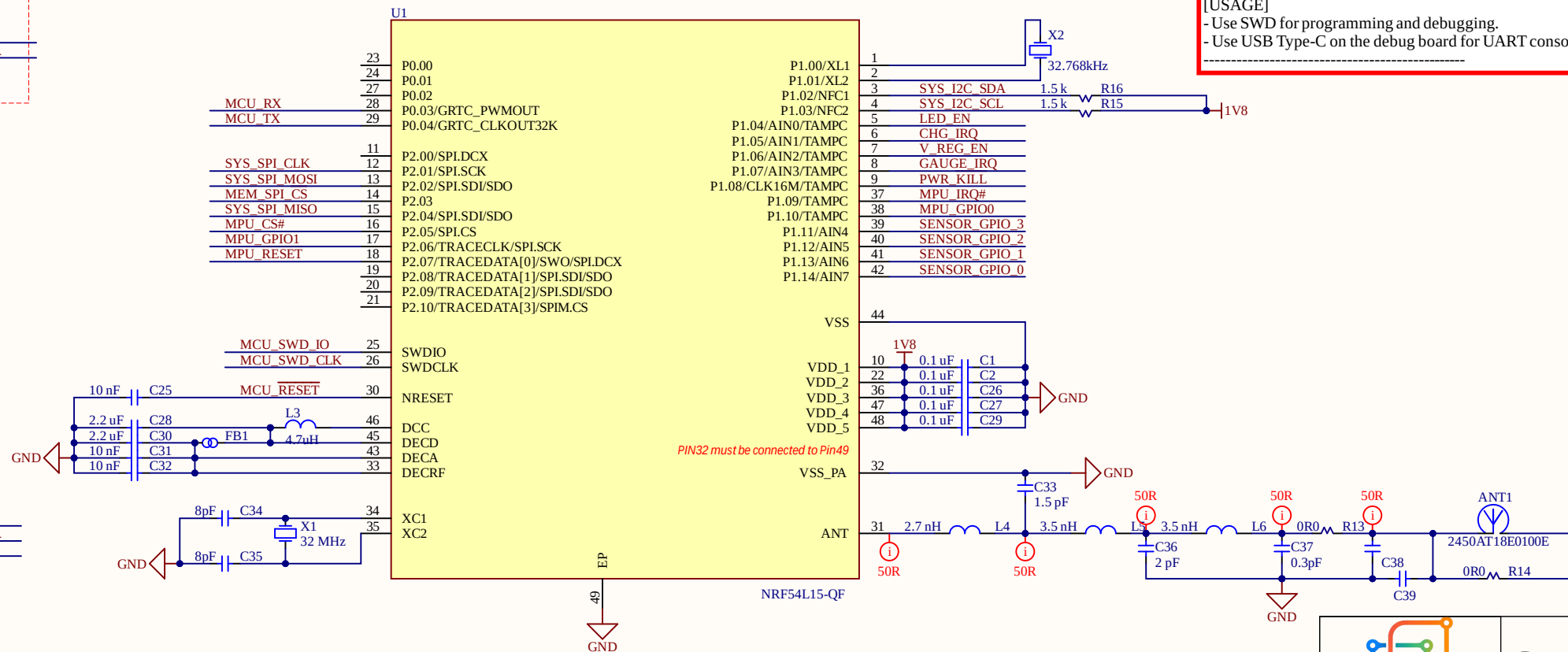
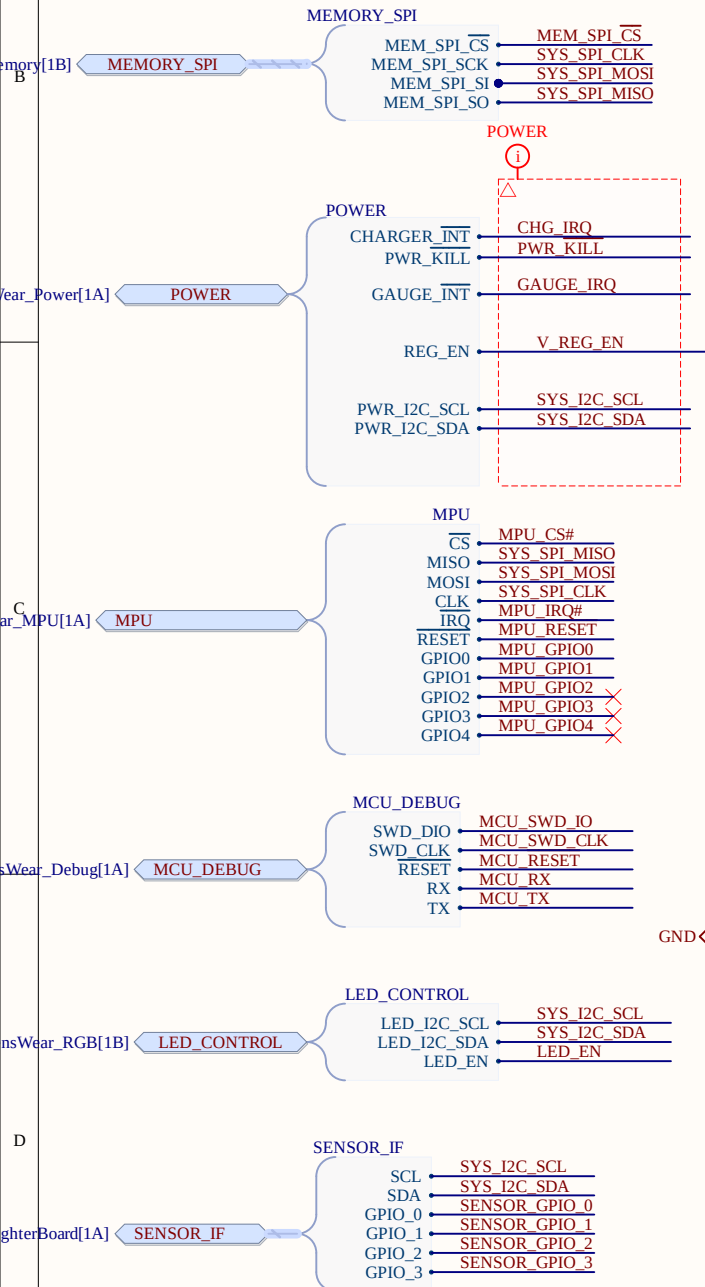
[DEBUG BOARD CONNECTION]

Connect the debug-board connector to:

- 1V8 / VREF
- GND
- SWDIO
- SWDCLK
- RESET#
- MCU_TX
- MCU_RX

[USAGE]

- Use SWD for programming and debugging.
- Use USB Type-C on the debug board for UART console/debug messages.



A

[1B]

B

C

D

1

2

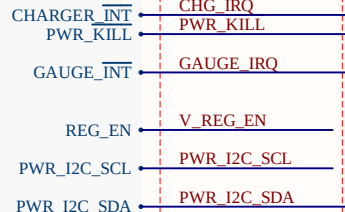
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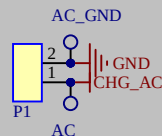
POWER



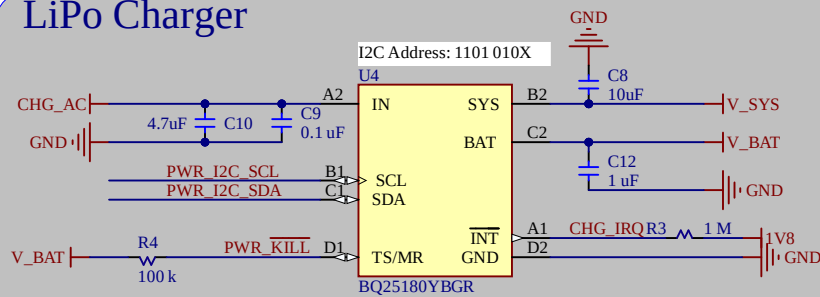
POWER



Power Supply



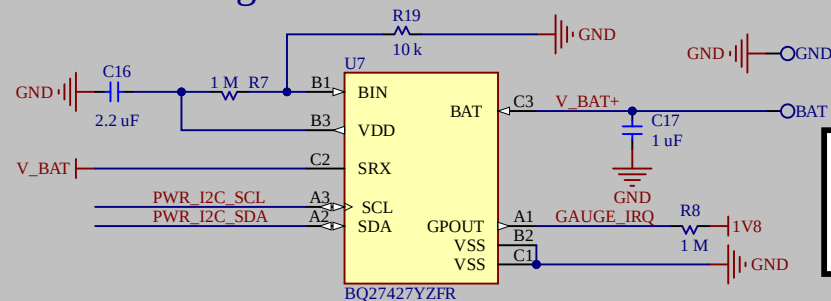
LiPo Charger



NO NTC is used in this design --
Rp = 1 M is required to allow different beta values of the NTC if we were to use one

1. @PWR_KILL can be used as long press based shutdown signal if it is needed.
2. CHG_AC must be connected to USB VBUS.
3. V_BAT is the supply from GAUGE, not BATTERY pack + terminal.

LiPo Gauge



BATTERY PACK must have
- VBAT+
- VBAT -
pins.

SYSTEM POWER SUBSYSTEM USAGE NOTES

Main Logic Level: 1.8V (Net: 1V8)
Control Bus: I2C21 (SDA: P1.02, SCL: P1.03)

[1. POWER RAIL ARCHITECTURE]

V_BAT : Raw LiPo battery voltage.
V_SYS : System power managed by Charger (U4).
Feeds the buck regulators.
1V8 : Main MCU/Logic supply (Fixed 1.8V via U3).
Always ON if V_SYS is present.
V_REG_IO : Auxiliary Sensor supply (via U6).
MCU controlled (see below).

[3. I2C DEVICE CONFIGURATION (Bus TWIM20)]

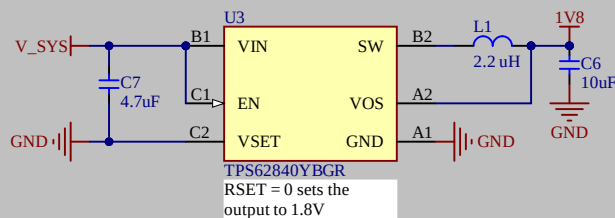
Device | Ref | Address (7-bit) | Notes

LiPo Charger | U4 | 0x6A (Fixed) | BQ25180. No NTC used.
LiPo Fuel Gauge | U7 | 0x55 (Fixed) | BQ27427. Low-side sensing.
Sensor Regulator | U6 | 0x68 (Fixed) | TPSM831021. Vout set via I2C.

[4. CRITICAL HARDWARE NOTES]

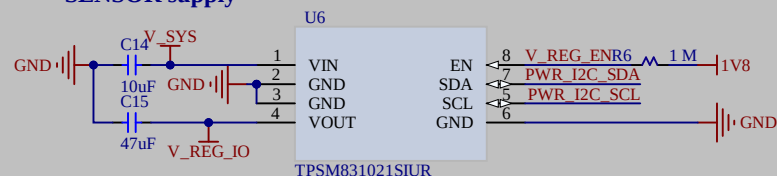
- BATTERY CONN : Design uses Low-Side sensing (R7).
Battery connector MUST have separate
PACK- and PACK+ terminals routed to U7.
- NTC THERM : Not used in this design. Charger configured
to ignore TS pin (check R4/R1 resistors).

SYS Supply



TPS62840YBGR
RSET = 0 sets the
output to 1.8V

SENSOR supply



[2. CONTROL & INTERRUPTS - MCU CONNECTIONS]

PWR_KILL (P1.08) : System Shutdown / Ship Mode Control.
Connects to Charger (U4).
- Drive LOW to cut V_SYS output (Power Off).
- Drive HIGH to keep system active.

V_REG_EN (P1.06) : Sensor Regulator (U6) Enable.
- HIGH = Output ON.
- LOW = Output OFF.

CHG_IRQ (P1.05) : Charger Interrupt (Active Low, Open Drain).
Signals charging status changes, faults, etc.

GAUGE_IRQ (P1.07) : Fuel Gauge Interrupt (Active Low, Open Drain).
Signals low battery, state of charge changes.



SensWear-V1R1

Designer/Engineer Yusein Ali	Board Title SensWear-Hub
Drawn By Yusein Ali	Document Name SensWear_Power.SchDoc
Approved By Dariush Salami	Rev. B0 Sheet 4 of 8 Date 01.03.2026

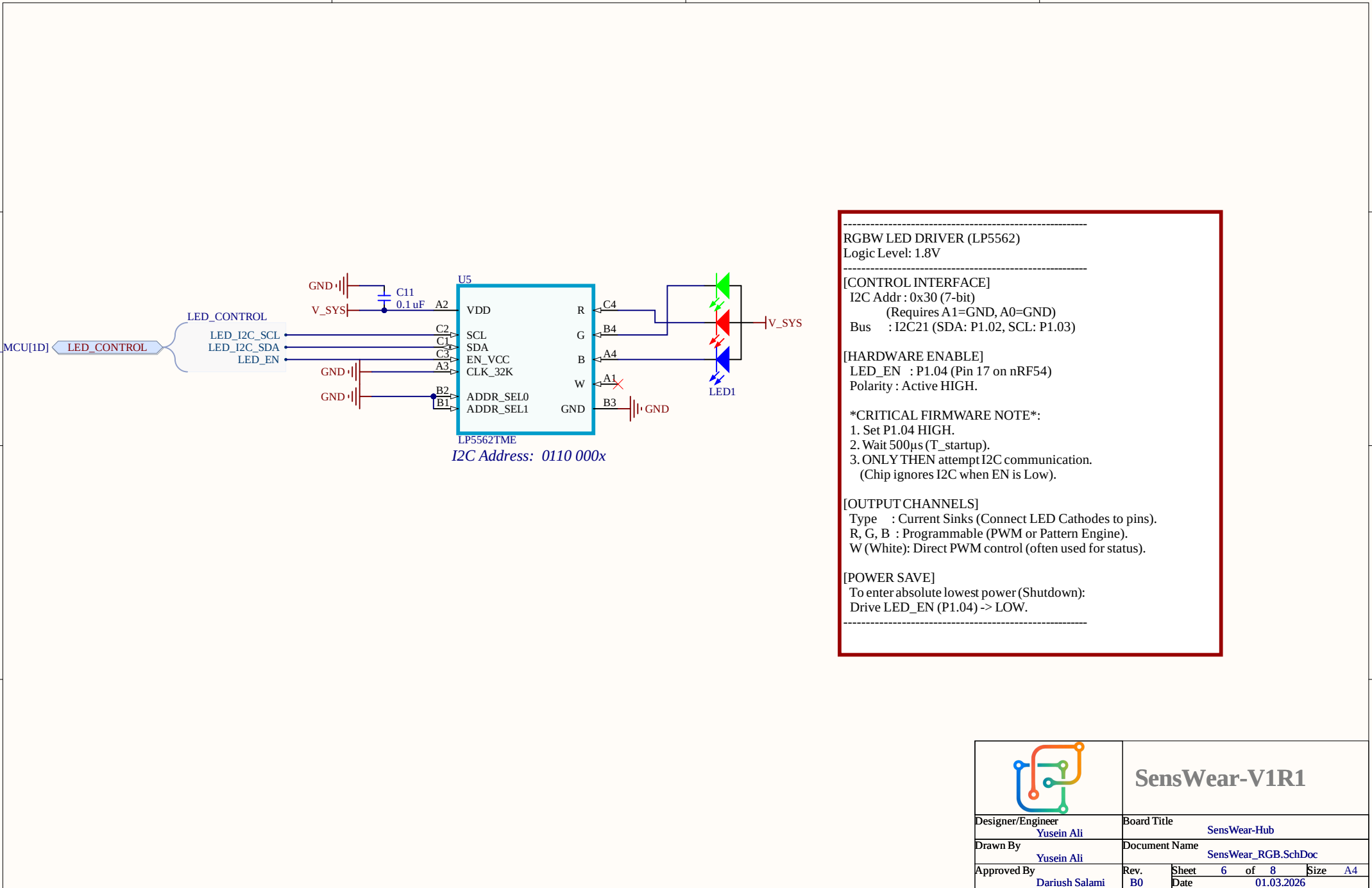
1

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3

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		SensWear-V1R1			
Designer/Engineer Yusein Ali		Board Title SensWear-Hub			
Drawn By Yusein Ali		Document Name SensWear_RGB.SchDoc			
Approved By Dariush Salami		Rev. B0	Sheet 6	of 8	Size A4
		Date	01.03.2026		

A

B

C

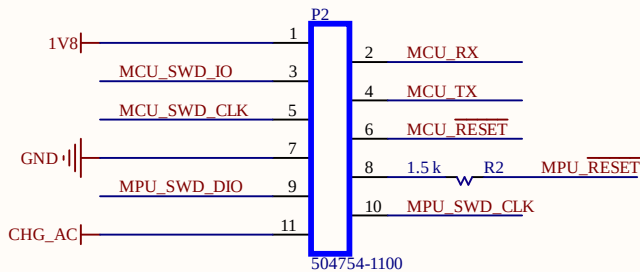
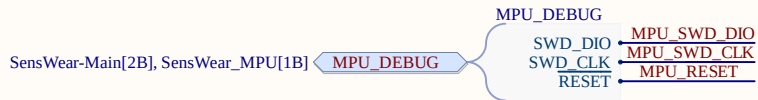
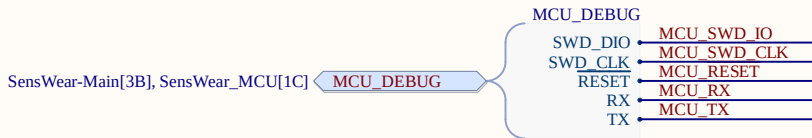
D

A

B

C

D



RESET LINE PROTECTION (MPU_RESET)

Component : Resistor (Series)

Value : 1.5kΩ

Location : Series between nRF54 Pin P2.07 and the Debug Connector (Pin 8).

Function : - Prevents short-circuit damage if nRF54 accidentally drives line HIGH while Debugger drives LOW.
- Current limited to ~1.2mA.

MPU DEBUG INTERFACE (BHI360)

Logic Level: 1.8V (VDDIO)

[JTAG / SWD CONNECTION] - Connector P2

MPU_SWD_DIO : Pin 9

- Connected to BHI360 Pin 16 (JTAG_DIO)

- Bi-directional Data

MPU_SWD_CLK : Pin 10

- Connected to BHI360 Pin 15 (JTAG_CLK)

- Clock Input

MPU_RESET# : Pin 8

- Active Low Reset

- Connected to BHI360 Pin 17 (RESETN)

[CRITICAL RESET NOTE]

The line 'MPU_RESET' is shared between:

1. External Debug Connector (Pin 8)
2. nRF54 MCU Pin P2.07 (Pin 7 on U1)

WARNING: To debug the BHI360 via Connector P2, the nRF54 MCU must configure P2.07 as INPUT (High-Z) to avoid driving the line HIGH while the debugger tries to pull it LOW.

[DEBUGGER COMPATIBILITY]

Target Core : ARC EM4 (Fuser Core)

Interface : 2-Wire JTAG/SWD

MCU (nRF54L15) DEBUG & DIAGNOSTIC INTERFACE

Logic Level: 1.8V (VDD_IO)

[SWD PROGRAMMING]

SWDIO : SWDIO (Physical Pin 25)

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- Has internal pull-up.

SWDCLK : SWDCLK (Physical Pin 26)

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- Has internal pull-down.

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[UART CONSOLE] - Instance: UART30

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MCU_TX : P0.04 (Pin 29)

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NOTE:

P0.03 also has GRTC PWM functionality.

P0.04 also has GRTC CLKOUT32K functionality.

Do not enable those GRTC functions while using these pins for UART.

[LAYOUT REQUIREMENTS]

1. Debug connector VREF/VTref must be connected to 1V8.

2. Debug probe / USB-UART must support 1.8V target I/O.

3. Do not place large capacitors on SWDIO or SWDCLK.

4. Keep SWDIO, SWDCLK, RESET#, UART_TX, and UART_RX short and clean.

5. UART_TX and UART_RX are single-ended signals, not differential pairs.

6. Provide a solid common GND between the debug board and the target board.

[DEBUG BOARD CONNECTION]

Connect the debug-board connector to:

- 1V8 / VREF

- GND

- SWDIO

- SWDCLK

- RESET#

- MCU_TX

- MCU_RX

[USAGE]

- Use SWD for programming and debugging.

- Use USB Type-C on the debug board for UART console/debug messages.



SensWear-V1R1

Designer/Engineer

Yusein Ali

Board Title

SensWear-Hub

Drawn By

Yusein Ali

Document Name

SensWear_Debug.SchDoc

Approved By

Dariush Salami

Rev.

B0

Sheet

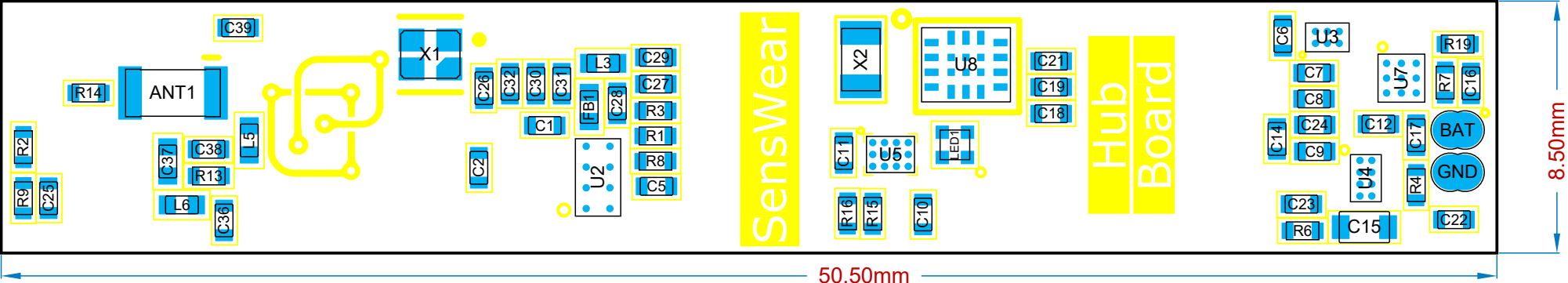
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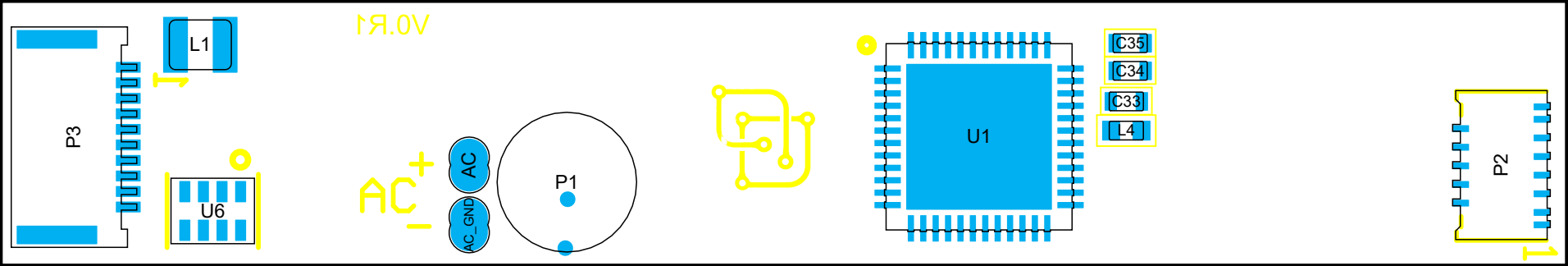
01.03.2026

of 8 Size A4

View from Top side (Scale 5:1)



View from Bottom side (Scale 5:1)



Bill Of Materials

Line #	Designator	Comment	Quantity
1	ANT1	2450AT18E0100E	1
2	C1, C2, C5, C9, C11, C18, C19, C26, C27, C29	0.1 uF	10
3	C6, C8, C14	10uF	3
4	C7, C10, C22, C23, C24	4.7uF	5
5	C12, C17	1 uF	2
6	C15	47uF	1
7	C16, C28, C30	2.2 uF	3
8	C21	220nF	1
9	C25, C31, C32	10 nF	3
10	C33	1.5 pF	1
11	C34, C35	8pF	2
12	C36	2 pF	1
13	C37	0.3pF	1
14	C38, C39	NC	2
15	FB1	BLM15AG121SN1D	1
16	L1	2.2 uH	1
17	L3	4.7uH	1
18	L4	2.7 nH	1
19	L5, L6	3.5 nH	2

Bill Of Materials

Line #	Designator	Comment	Quantity
20	LED1	SML-LX0404SIUPGUSB	1
21	P1	RM-1238-B02-41	1
22	P2	504754-1100	1
23	P3	51281-1098	1
24	R1, R3, R6, R7, R8, R9	1 M	6
25	R2, R15, R16	1.5 k	3
26	R4	100 k	1
27	R13, R14	0R0	2
28	R19	10 k	1
29	U1	NRF54L15-QF	1
30	U2	M95P32-IXCST/EF	1
31	U3	TPS62840YBGR	1
32	U4	BQ25180YBGR	1
33	U5	LP5562TME	1
34	U6	TPSM831021SIUR	1
35	U7	BQ27427YZFR	1
36	U8	BHI360	1
37	X1	32 MHz	1
38	X2	32.768kHz	1

		SensWear-V1R1		
Designer/Engineer		Board Title		
Yusein Ali		SensWear-Hub		
Drawn By		Document Name		
Yusein Ali		SensWear-Hub-assembly.PCBDwf		
Approved By		Ver.	Rev.	Date
Dariush Salami		1	B0	01.03.2026